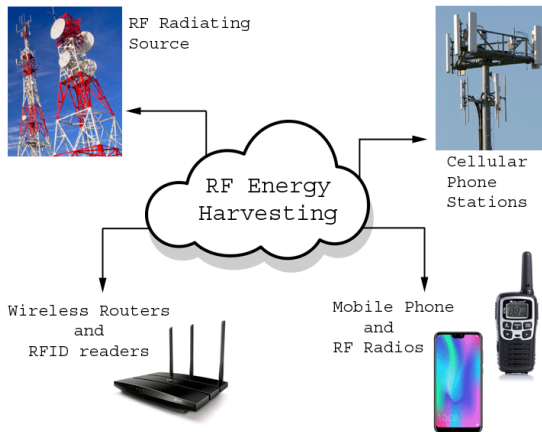


Internet das Coisas e Redes Veiculares (TP-546)

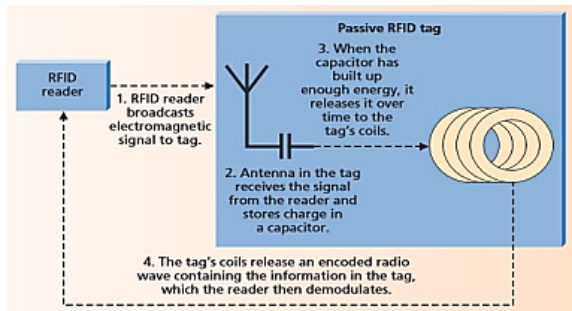
Samuel Baraldi Mafra

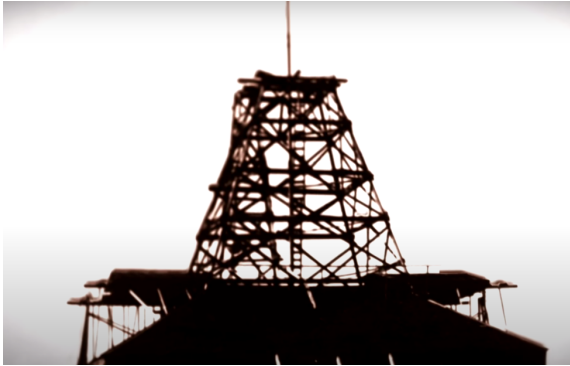


Captação de energia de sinais de radio frequência:



RFID

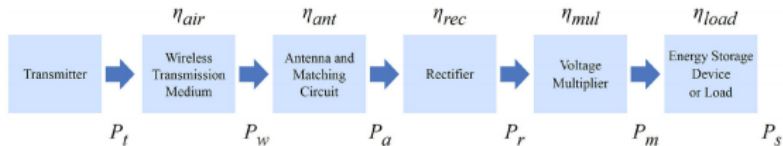




Tesla Conducts Impressive Wireless Electricity Experiments

<https://youtu.be/q0i67E480V0>

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Captação de energia de sinais ISDB-T

R. Shigeta et al., "Ambient RF Energy Harvesting Sensor Device With Capacitor-Leakage-Aware Duty Cycle Control," in IEEE Sensors Journal, vol. 13, no. 8, pp. 2973-2983, Aug. 2013

R. Vyas, H. Nishimoto, M. Tentzeris, Y. Kawahara and T. Asami, "A battery-less, energy harvesting device for long range scavenging of wireless power from terrestrial TV broadcasts," 2012 IEEE/MTT-S International Microwave Symposium Digest, Montreal, QC, 2012, pp. 1-3

- A colheita de energia através de sinais de tv é uma boa opção, por se tratar de uma banda onde quase todo o tempo há sinais trafegando e também pelo fato de estar presente em praticamente todos os ambientes.
- Como pontos negativos, podemos citar o fato de a fonte de transmissão, na maioria das vezes, ficar longe do receptor, visto que necessita-se de poucos transmissores para cobrir uma extensa área e também o tamanho da antena.

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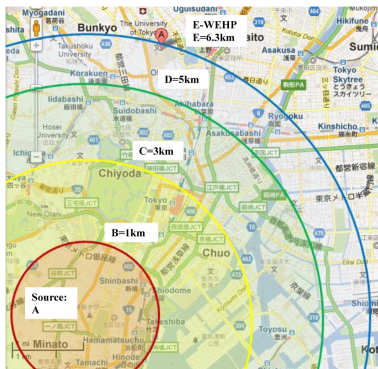
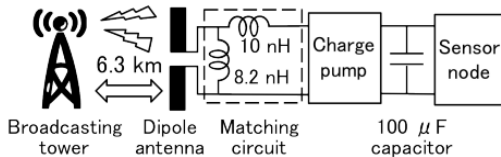


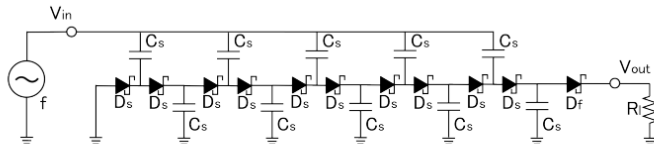
TABLE II
THEORETICAL AND MEASURED SINGLE-CARRIER AND CHANNEL POWER LEVELS USING E-WEHP ANTENNA (GAIN: 5–7.3 dBi)

TV Channel Freq. (MHz)		A P _{TX} Power kW	B 1km μW	C 3km μW	D 5km μW	E 6.3km μW	
						theo ry	meas.
JOMX 512-518	1-tone	5	0.32	0.26	0.21	0.19	0.008
	Chan nel		37.3	30.3	24.5	22.1	0.93
JO- RX/CX/TX/ EX/AX/AB /AK 518-560	1-tone	48	3.6- 4.1	2.9- 3.4	2.3- 2.7	2.2- 2.5	0.1- 0.25
	Chan nel		419- 478	338- 396	268- 315	256- 291	11.7- 29
JODJ 560-566	1-tone	19	1.37	1.08	0.86	0.78	0.032
	Chan nel		160	126	100	91	3.73

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Retificador e multiplicador de tensão

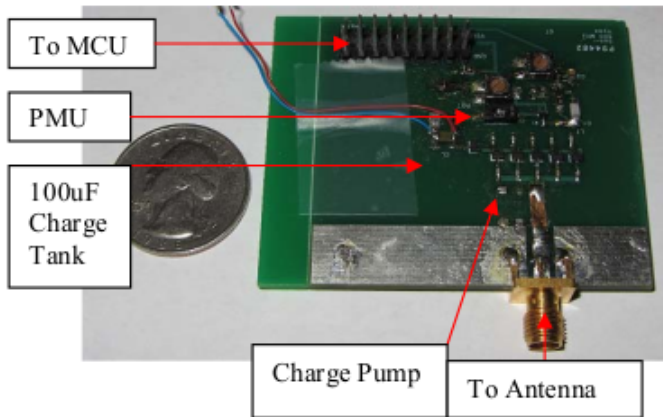


Input Frequency f : 540MHz, Number of Stages: 5

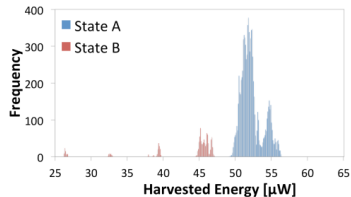
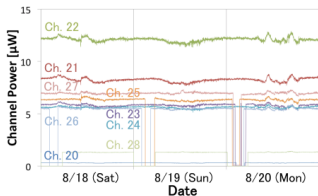
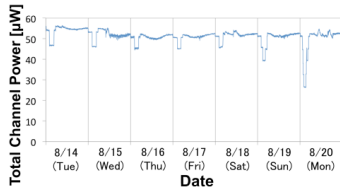
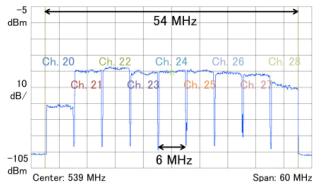
Stage Capacitor C_s : 15pF, Stage Diode D_s : Skyworks Solutions SMS-7630

Final Diode D_f : Avago HSMS-286x, Load R_l : 470k Ω

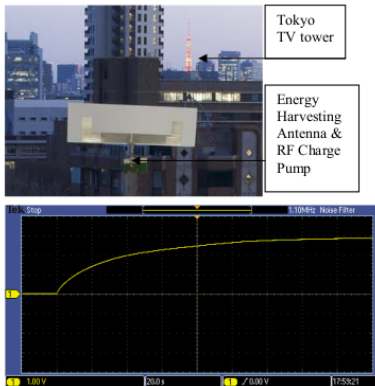
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Captação de sinais de WI-FI:

Y. Chen and C. Chiu, "Maximum Achievable Power Conversion Efficiency Obtained Through an Optimized Rectenna Structure for RF Energy Harvesting," in IEEE Transactions on Antennas and Propagation, vol. 65, no. 5, pp. 2305-2317, May 2017

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- A colheita de energia derivada da faixa de frequência Wi-Fi é uma boa opção quando se trata de aplicações que necessitam de mobilidade e escalabilidade;
- Isto se dá devido a antena para esta faixa ser muito pequena, ainda mais quando comparada com uma antena para colheita de sinais de tv digital;



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- Presente em ambientes urbanos, especialmente em áreas residenciais;
- Esta banda pode ser facilmente usada para colheita de energia e para comunicação, sendo possível realizar projetos que realizem as duas tarefas.
- Por outro lado, devido as baixas potências de entrada, somente pode ser utilizado para aplicações de baixo consumo de energia.



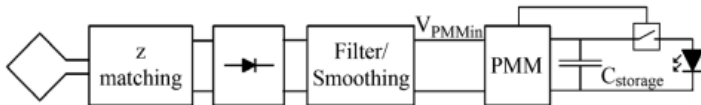
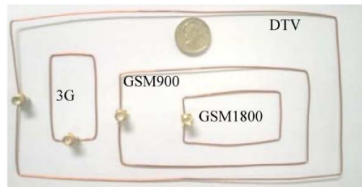
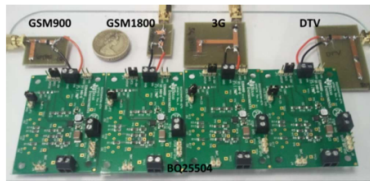
Captar de múltiplas frequências:

M. Piñuela, P. D. Mitcheson and S. Lucyszyn, "Ambient RF Energy Harvesting in Urban and Semi-Urban Environments," in IEEE Transactions on Microwave Theory and Techniques, vol. 61, no. 7, pp. 2715-2726, July 2013

V. Kuhn, C. Lahuec, F. Seguin and C. Person, "A Multi-Band Stacked RF Energy Harvester With RF-to-DC Efficiency Up to 84%," in IEEE Transactions on Microwave Theory and Techniques, vol. 63, no. 5, pp. 1768-1778, May 2015.

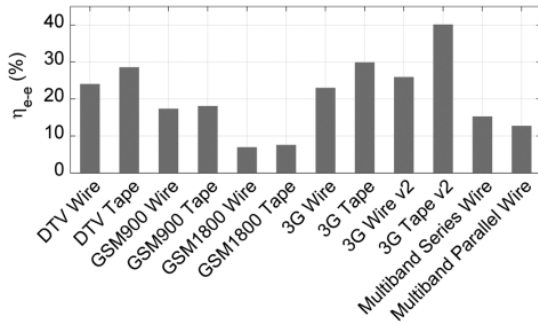
- Sinais multibandas concedem um resultado superior aos demais, devido ao fato de se obter sinais de diferentes frequências. Para ambientes que dispõem de diferentes sinais trafegando é adequado que se use esse método, para obter o máximo de sinais possíveis.

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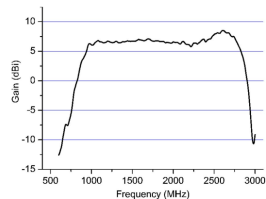
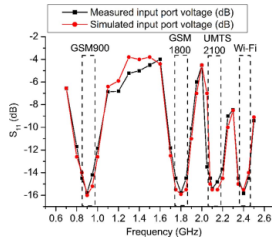
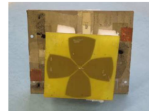
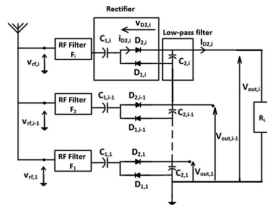


M. Piñuela, P. D. Mitcheson and S. Lucyszyn, "Ambient RF Energy Harvesting in Urban and Semi-Urban Environments," in IEEE Transactions on Microwave Theory and Techniques, vol. 61, no. 7, pp. 2715-2726, July 2013

Internet das Coisas e Redes Veiculares

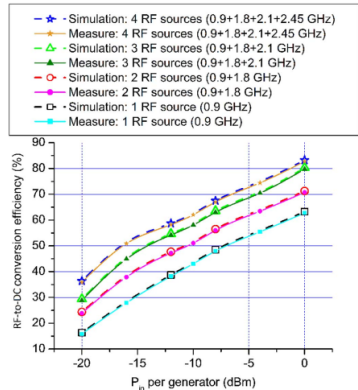
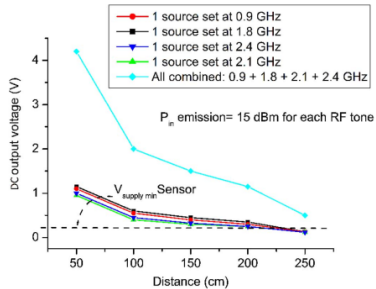


Internet das Coisas e Redes Veiculares



V. Kuhn, C. Lahuec, F. Seguin and C. Person, "A Multi-Band Stacked RF Energy Harvester With RF-to-DC Efficiency Up to 84%," in IEEE Transactions on Microwave Theory and Techniques, vol. 63, no. 5, pp. 1768-1778, May 2015.

Internet das Coisas e Redes Veiculares



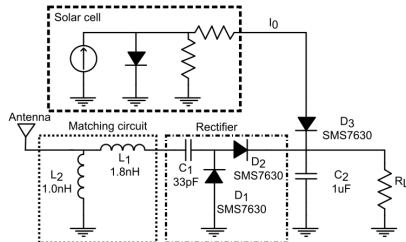
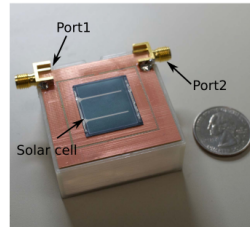
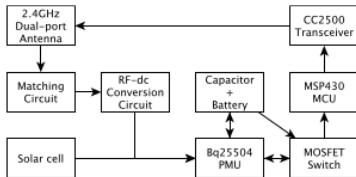
Solução Híbrida Solar-RF:

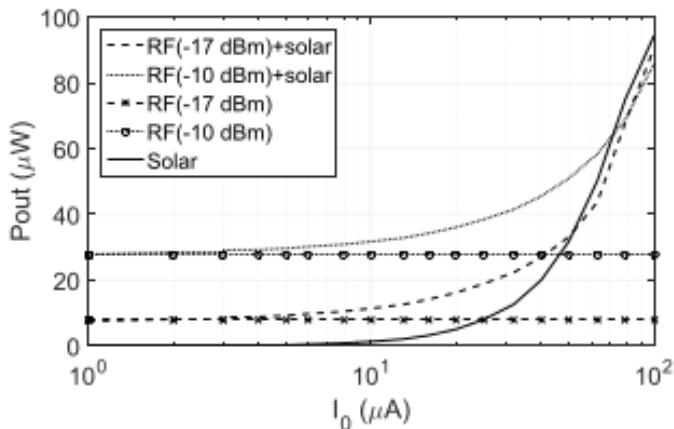
A captação de energia por sinais de RF gera potências muito baixas, é necessário uma alternativa para conseguir melhores resultados, sendo essa, a combinação com outras fontes.

J. Bito, R. Bahr, J. G. Hester, S. A. Nauroze, A. Georgiadis and M. M. Tentzeris, "A Novel Solar and Electromagnetic Energy Harvesting System With a 3-D Printed Package for Energy Efficient Internet-of-Things Wireless Sensors," in IEEE Transactions on Microwave Theory and Techniques, vol. 65, no. 5, pp. 1831-1842, May 2017

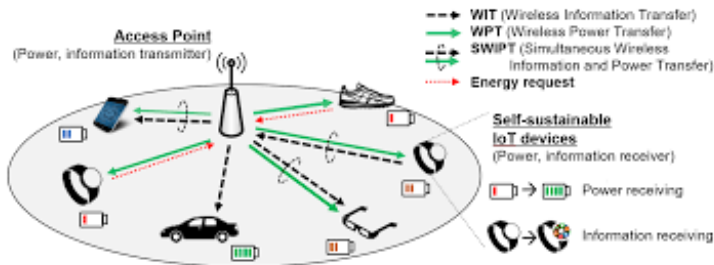
- Uma alternativa para aumentar a energia final é combinando os diversos tipos de energia. Para aplicações onde se tem unidades de armazenamento de energia é interessante o uso dessa tecnologia, pois a energia adquirida consegue manter a bateria carregada enquanto ela alimenta um circuito.

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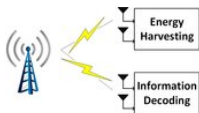




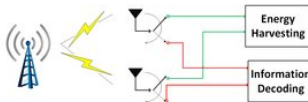
Técnicas de transferência simultânea de energia e informação.



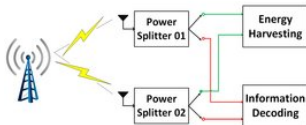
Técnicas de transferência simultânea de energia e informação.



a) Separate Receiver Architecture



b) Time Switching Architecture

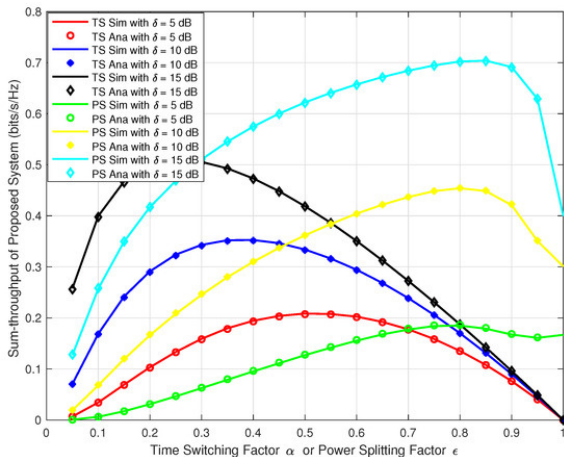


c) Power Splitting Architecture

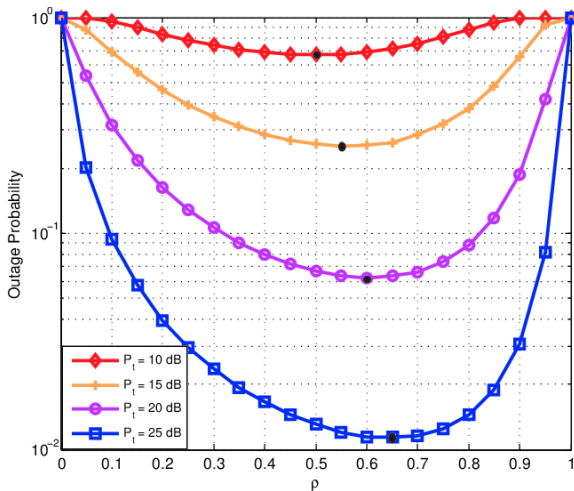


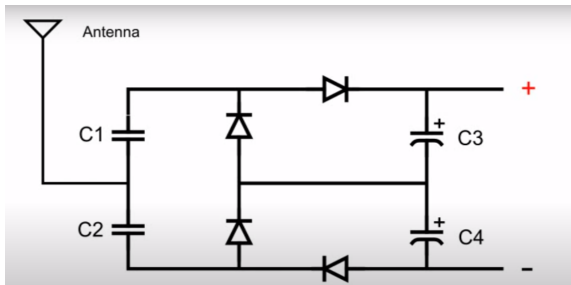
d) Antenna Switching Architecture

Time switching-Power splitting



Power splitting





Energy Harvesting from Electromagnetic Signals - Rectenna
<https://www.youtube.com/watch?v=XpLCK88nVgU>



Energy Harvesting
with Tom Sparkman

Energy Harvesting Demonstration

<https://www.youtube.com/watch?v=C7G7TyNL01Q>

Como manter um nó receptor em modo ultra baixo consumo e ao mesmo tempo deixar o circuito receptor ligado?

Wake-up Radios:

Os rádios Wake-up são geralmente projetados como um receptor de rádio secundário de potência quase zero, usado para acionar o rádio principal quando uma nova comunicação é iniciada.



IEEE Wake-Up Radio

Prolong the battery life of Internet of Things devices with this low-power, high-performance solution.

The Problem:

Today, wireless networked devices have to enter a sleep state to prolong their battery life. The longer the device sleeps, the longer the battery lasts, but the lower the device performance. Low power consumption and high performance are conflicting goals.

The Solution:

IEEE Wake-Up Radio from IEEE 802.11ba standards task group lets devices achieve low power and high performance (low latency) AT THE SAME TIME!

How it Works



1 A low power radio receiver (Wake-Up Radio) is added to a device. **This is the only thing that stays "awake" all the time**, and it uses very little battery power.



2 The low power radio "listens" for a special signal that says that information is being sent to the device.



3 The radio "wakes up" the main Wi-Fi® radio so that data exchange can begin.

Wake-Up Radio Applications

Many Internet of Things devices that run over short-range wireless networks can benefit from IEEE Wake-Up Radio, including:

Transportation and Logistics



Smart Warehouses



Smart Homes



Agriculture



Health Monitoring Wearable Devices



Get the information you need to begin developing low-power, high-performance Internet of Things devices with the newest technology report from IEEE: IEEE Technology Report on Wake-Up Radio: An Application, Market, and Technology Impact Analysis of Low-Power/Low-Latency 802.11 Wireless LAN Interfaces

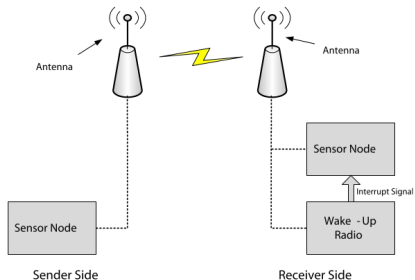


Picoradios:

- Tamanho e peso reduzidos;
- Baixo consumo;
- Captar energia do meio.
- Brasileiro contribuiu na pesquisa durante o pós doc na Universidade da California, Berkeley: Julio Leão da Silva Jr.

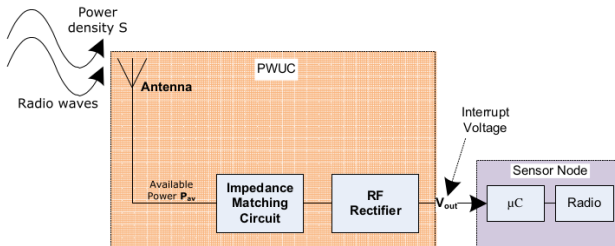
J. Rabaey, J. Ammer, J. L. da Silva and D. Patel, "PicoRadio: Ad-hoc wireless networking of ubiquitous low-energy sensor/monitor nodes," Proceedings IEEE Computer Society Workshop on VLSI 2000. System Design for a System-on-Chip Era, Orlando, FL, USA, 2000, pp. 9-12,

Wake-up radio

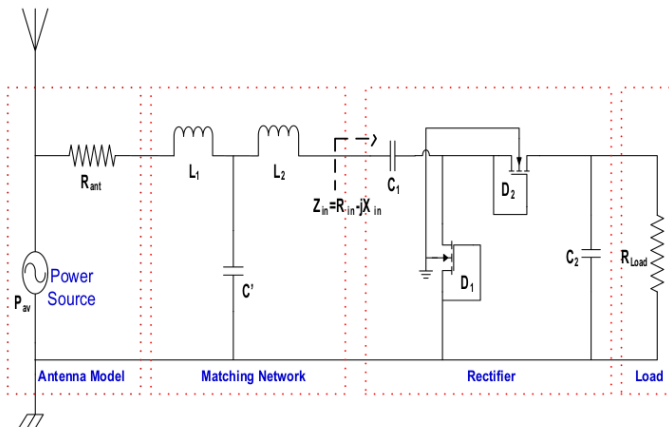


A Passive Wake-Up Circuit for Event Driven Wireless Sensor Network Applications Chandra Shekhar (Microelectronics Division, IIIT Allahabad, India), Shirshu Varma (IT Division, IIIT Allahabad, India), and M. Radhakrishna (Microelectronics Division, IIIT Allahabad, India) Journal of Circuits, Systems and Computers 2015 24:08

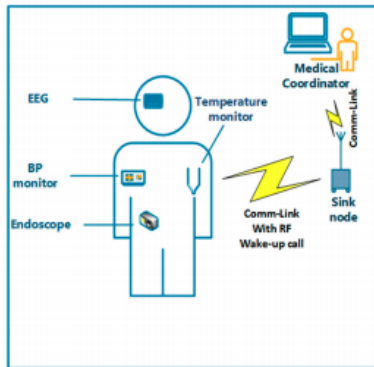
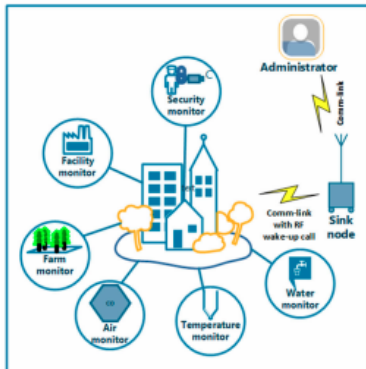
Wake-up radio



Wake-up radio: Circuito

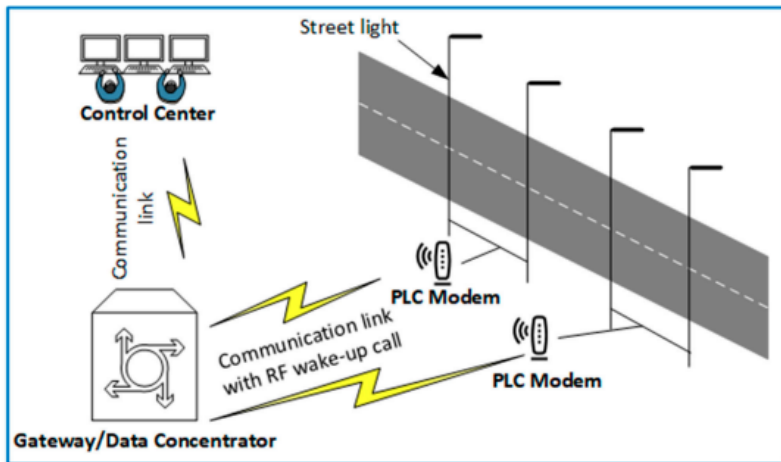


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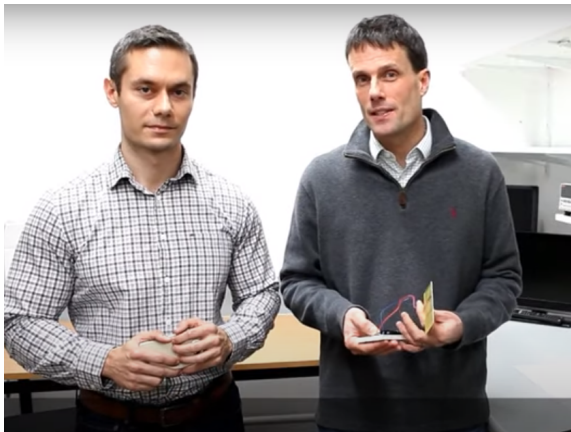


Bello, H.; Xiaoping, Z.; Nordin, R.; Xin, J. Advances and Opportunities in Passive Wake-Up Radios with Wireless Energy Harvesting for the Internet of Things Applications. *Sensors* 2019, 19, 3078.

Internet das Coisas e Redes Veiculares



Bello, H.; Xiaoping, Z.; Nordin, R.; Xin, J. Advances and Opportunities in Passive Wake-Up Radios with Wireless Energy Harvesting for the Internet of Things Applications. *Sensors* 2019, 19, 3078.



Zero-power circuit that listens for radio waves

<https://www.youtube.com/watch?v=Z0zvw6YIc7U>



Drone in Mid-Flight Uses Radio Waves to Recharge a Sensor on the Ground

https://www.youtube.com/watch?v=fX2tt7etBDw&feature=emb_logo

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EnOcean



Batteryless wireless switches control light and shading



Batteryless wireless switches perform remote-control functions



Occupancy sensors trigger lighting and climate control systems



Wireless connectors control and monitor household appliances



Networked smoke sensors set off fire alarms to trigger emergency response



Batteryless wireless window contacts monitor window status



Batteryless wireless door/window handles monitor door/window status



Batteryless wireless control units allow for optimal climate conditions and maximum operating comfort



Wireless actuators control radiators, room controllers govern underfloor heating



Sunblind actuators control the sunshade elements

EnOcean Alliance - the wireless standard for sustainable buildings

<https://www.youtube.com/watch?v=YKY06ns1i4o&t=3s>